

Supplementary: Your Classifier can Secretly Suffice

Multi-Source Domain Adaptation

Anonymous Author(s)

Section	Content
Sec. 1	Results on <i>DomainNet</i>
Sec. 2	Implicit alignment under Category-shift
Sec. 3	Practical Application: Cross-Domain Image Retrieval
Sec. 4	Code Reference

1 Results on DomainNet

Owing to the limits of space, we present a summary of results on *DomainNet* in the paper. The full comparison is presented here in Table 1. The results for the prior arts are reported from [4].

Table 1: Results on the *DomainNet* [4] dataset.

	Method	→Clp	→Inf	→Pnt	→Qdr	→Rel	→Skt	Avg
SB	Source Only	39.6±0.5	8.2±0.7	33.9±0.6	11.8±0.6	41.6±0.8	23.1±0.7	26.4±0.7
	DAN [1]	39.1±0.5	11.4±0.8	33.3±0.6	16.2±0.3	42.1±0.7	29.7±0.9	28.6±0.6
	RTN [3]	35.3±0.7	10.7±0.6	31.7±0.8	13.1±0.6	40.6±0.5	26.5±0.7	26.3±0.7
	JAN [2]	35.3±0.7	9.1±0.6	32.5±0.6	14.3±0.6	43.1±0.7	25.7±0.6	26.7±0.6
	ADDA [6]	39.5±0.8	14.5±0.6	29.1±0.7	14.9±0.5	41.9±0.8	30.7±0.6	28.4±0.7
	MCD [5]	42.6±0.3	19.6±0.7	42.6±0.9	3.8±0.6	50.5±0.4	33.8±0.8	32.2±0.6
SC	Source Only	47.6±0.5	13.0±0.4	38.1±0.4	13.3±0.3	51.9±0.8	33.7±0.5	32.9±0.5
	DAN [1]	45.4±0.4	12.8±0.8	36.2±0.5	15.3±0.3	48.6±0.7	34.0±0.5	32.1±0.5
	RTN [3]	44.2±0.5	12.6±0.7	35.3±0.5	14.6±0.7	48.4±0.6	31.7±0.7	31.1±0.6
	JAN [2]	40.9±0.4	11.1±0.6	35.4±0.5	12.1±0.6	45.8±0.5	32.3±0.6	29.6±0.5
	ADDA [6]	47.5±0.7	11.4±0.6	36.7±0.5	14.7±0.5	49.1±0.8	33.5±0.4	32.2±0.6
	MCD [5]	54.3±0.6	22.1±0.7	45.7±0.6	7.6±0.4	58.4±0.6	43.5±0.5	38.5±0.6
MS	DCTN [7]	48.6±0.7	23.5±0.5	48.8±0.6	7.2±0.4	53.5±0.5	47.3±0.4	38.2±0.5
	M ³ SDA [4]	57.2±0.9	24.2±1.2	51.6±0.4	5.2±0.4	61.6±0.8	49.6±0.5	41.5±0.7
	M ³ SDA-β [4]	58.6±0.5	26.0±0.8	52.3±0.5	6.3±0.5	62.7±0.5	49.5±0.7	42.6±0.6
	SImpAl ₁₀₁	66.4±0.8	26.5±0.5	56.6±0.7	18.9±0.8	68.0±0.5	55.5±0.3	48.6±0.6

2 Implicit Alignment under Category-shift

We find that SImpAl works well even under category-shift. In Fig. 4c of the paper, we compare the relative drop (%) in accuracy under category-shift which is defined as,

$$\text{relative drop} = \frac{(A_{\text{vanilla}} - A_{\text{category-shift}})}{A_{\text{vanilla}}} \times 100 \quad (1)$$

Our approach exhibits a lower drop in accuracy. To understand the phenomenon, we further investigate the latent space alignment of the domains after adaptation using SImpAl. We show two t-SNE plots, 1) Sources-only, *i.e.* showing alignment among the source domains, and 2) Sources + Target, *i.e.* showing the alignment among all sources and the target domains.

Fig. 1 shows the latent space alignment for the task **A, D, W**→**C** for the *Office-Caltech* dataset. For the **Overlap** scenario, we set the number of shared categories to 2, while the number of source-private categories are 3, 3, 2 for the sources Amazon (**A**), DSLR (**D**) and Webcam (**W**) respectively. Clearly, alignment among the sources is observed only for the two shared categories (annotated in Fig. 1).

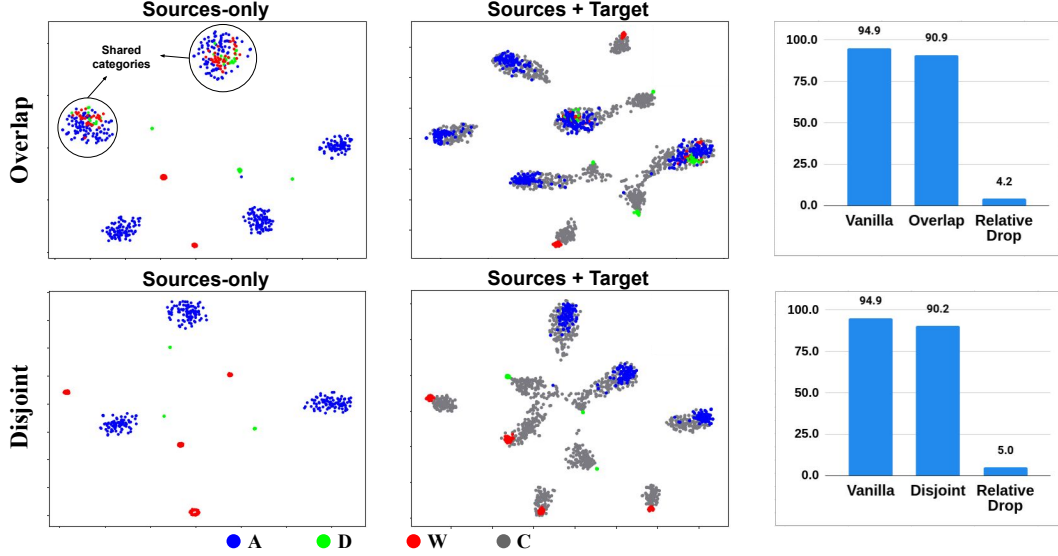


Figure 1: Pre-classifier space t-SNE showing the alignment of features under the category-shift settings **Overlap** (top) and **Disjoint** (bottom) on the *Office-Caltech* dataset for the task **A, D, W**→**C**. Note the alignment of selected clusters (shared classes among the source domains) in the **Overlap** scenario, while none of the source clusters align in the **Disjoint** scenario. Best viewed in color.

15 In contrast, for the **Disjoint** scenario, the source domains **A, D, W** contain 3, 3, 4 unique classes
 16 respectively. Here, none of the source clusters align as expected (since each source has a distinct set
 17 of classes). However, in both scenarios, we find that the target domain aligns with at least one source
 18 domain (multiple sources if categories are shared). We find a similar trend across other datasets as
 19 shown in Fig. 2 for the **I, P**→**C** task of *ImageCLEF* (**Overlap**: 4 shared and 4, 4 source-private
 20 classes; **Disjoint**: 6, 6 source-private classes), and in Fig. 3 for the **A, D**→**W** task of *Office-31*
 21 (**Overlap**: 11 shared and 10, 10 source-private classes; **Disjoint**: 16, 15 source-private classes).

22 These observations demonstrate the success of SImpAI. While prior methods either directly or
 23 indirectly align all sources to a common semantic granularity [4, 7], our approach aligns only the
 24 shared categories. Therefore, especially under the category-shift, prior methods potentially align even
 25 the source-private categories among the source domains, while in SImpAI, such detrimental effects
 26 are not observed which explains the lower relative drop in performance.

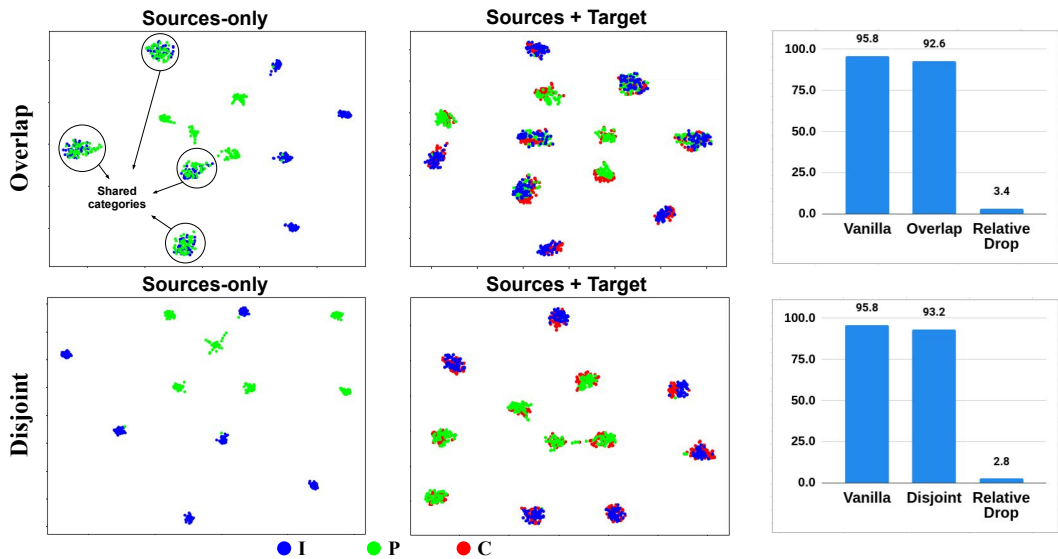


Figure 2: Pre-classifier space t-SNE showing the alignment of features under the category-shift settings **Overlap** (top) and **Disjoint** (bottom) on the *ImageCLEF* dataset for the task **I, P**→**C**. Best viewed in color.

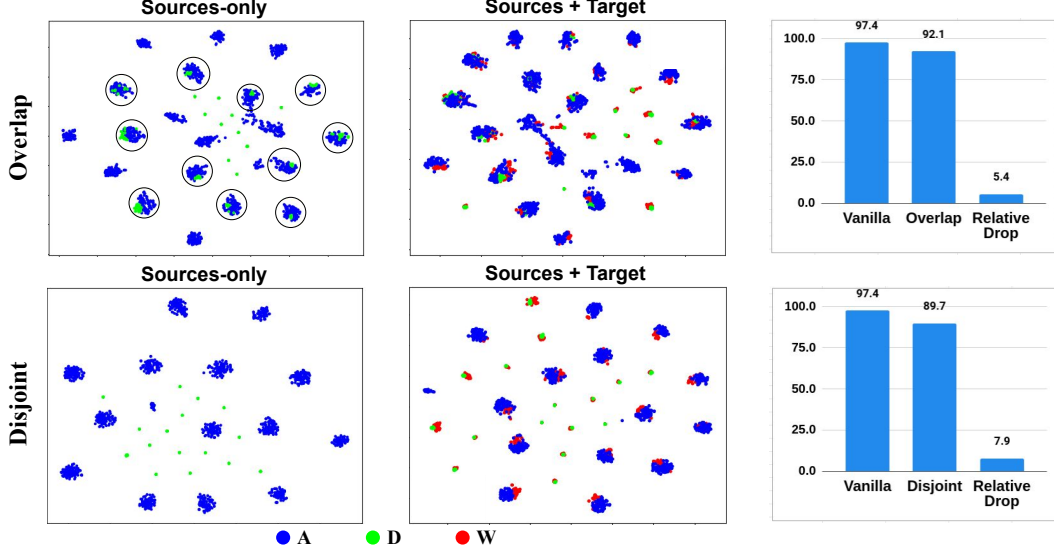


Figure 3: Pre-classifier space t-SNE showing the alignment of features under the category-shift settings **Overlap** (top) and **Disjoint** (bottom) on the *Office-31* dataset for the task **A, D** $\rightarrow$ **W** (in line with Fig. 4c of the paper). Source clusters corresponding to the shared classes are encircled. Best viewed in color.

3 Practical Application: Cross-Domain Image Retrieval

We now demonstrate a practical application of Implicit Alignment. Consider an unlabeled dataset $\mathcal{D}$ of images sampled from a target domain. Suppose we wish to retrieve images from this dataset based on some class semantics (for *e.g.*, "retrieve images of objects having wheels"). With an unannotated dataset, this task seems non-trivial. However, we show that this is possible by performing Multi-Source Domain Adaptation on the target dataset $\mathcal{D}$ using SImpAI. Considering $\mathcal{D}$ as the target domain, we adapt a deep model using labeled source datasets and the unlabeled target data $\mathcal{D}$, under the SImpAI framework, and use this model to measure semantic similarity.

We develop a cross-domain image retrieval system where, given a reference image $\mathbf{x}_r$, we retrieve semantically similar images from a given set of images $\mathcal{D}$ (called the query set). To achieve this, we obtain the nearest neighbors of the reference image $\mathbf{x}_r$ in the set $\mathcal{D}$, based on the class probability vector $\mathbf{p}$ (as obtained in Eq. 1 of the paper). This is shown in Algorithm 1 below.

Algorithm 1 Cross-Domain Image Retrieval

- 1: **require:** Reference image $\mathbf{x}_r$, Query set $\mathcal{D}$, Model $\{f, h\}$ ▷ $\cup$ denotes union
 - ▷ Calculate the class probability vector for $\mathbf{x}_r$ (using Eq. 1 of the paper)
 - 2: $\mathbf{M} \leftarrow h \circ f(\mathbf{x}_r)$
 - 3: $\mathbf{p}_r \leftarrow \frac{1}{n_d} \sum_{i=1}^{n_d} \sigma(\mathbf{M}_{[i:]})$
 - ▷ Calculate class probability vectors for images in the query set $\mathcal{D}$ (using Eq. 1 of the paper)
 - 4: $\mathcal{R} \leftarrow \{\}$ ▷ Create an empty collection of distance values
 - 5: **for** $\mathbf{x}_q \in \mathcal{D}$ **do**
 - 6: $\mathbf{M} \leftarrow h \circ f(\mathbf{x}_q)$
 - 7: $\mathbf{p}_q \leftarrow \frac{1}{n_d} \sum_{i=1}^{n_d} \sigma(\mathbf{M}_{[i:]})$
 - 8: $\mathcal{R} \leftarrow \mathcal{R} \cup \{\|\mathbf{p}_q - \mathbf{p}_r\|_2\}$ ▷ get the euclidean distance in class probabilities
 - 9: $\mathcal{D}' \leftarrow$ set of k nearest images in the query set $\mathcal{D}$ based on the distance $\mathcal{R}$
 - 10: **return** $\mathcal{D}'$
-

Note, Algo. 1 does not require any label information. Thus, the query set $\mathcal{D}$ can be unlabeled. We show an example use-case where we consider images in the "Quickdraw" (**Qdr**) domain of *DomainNet* [4] as the reference images, while the query set is the unlabeled target domain "Real" (**Rel**). Fig. 4 shows sample images in the two domains.

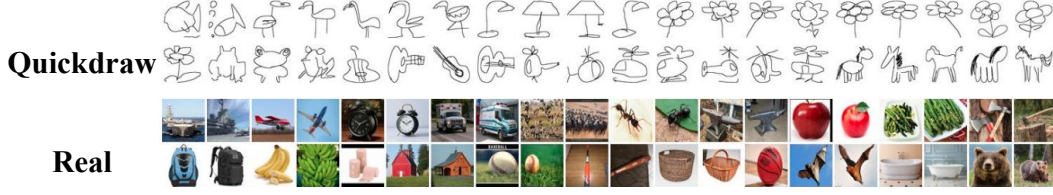


Figure 4: Sample images in the Quickdraw (**Qdr**) and Real (**Rel**) domains of the *DomainNet* [4] dataset. Note, this figure shows example images in the dataset; the images do not correspond.

We obtain a model adapted to the "Real" target domain (*i.e.* the model corresponding to $\rightarrow$ **Rel** in Table 1). Given a reference image in the **Qdr** domain, an end-user can retrieve semantically similar images from the unlabeled **Rel** domain using Algo. 1 above.

In Fig. 5, we present qualitative results, demonstrating the retrieval of images from the **Rel** domain using randomly selected images from the **Qdr** domain as reference. Note that, all images in Fig. 5 pertain to the test set of the corresponding domains that the model has not encountered during adaptation. The retrieval process using Algo. 1 above works surprisingly well, yielding highly satisfactory results (annotated with green tick-mark). While there are certain images which yield false retrievals (annotated with red cross-mark), many of those cases have incomprehensible reference image (marked with an orange question-mark) and can be ignored during qualitative evaluation.

Further, we design a tool to retrieve images from a chosen query set, by manually "doodling" class semantics (similar to the images in the **Qdr** domain). This tool is bundled with the code submission (See code reference in Sec. 4). **A video demonstration is provided in the Supplementary material (see image_retrieval_demo.mp4).** Fig. 6 shows the images retrieved during the demonstration.



Figure 5: Images retrieved from the Real (**Rel**) domain (below) using reference images from the Quickdraw (**Qdr**) domain (above). The model corresponds to the $\rightarrow$ **Rel** task of *DomainNet*. Here, we show the nearest neighbor ($k = 1$ in Line 9 of Algo. 1 above). All images pertain to the test set of *DomainNet*. Green tick-marks indicate satisfactory retrievals, red cross-marks indicate false retrievals and orange question-marks indicate the cases where the reference image is incomprehensible. Note the high success rate of retrievals.

4 Code Reference

We summarize the references to the code corresponding to each experiment performed in the paper. Please refer to the README.md file for further instructions.

- Main Results (Table 1 of the paper): code/train_code
- Category-Shift (Fig. 4c of the paper): code/union_disjoint_code
- Cross-domain Image Retrieval Tool (requires trained model): code/img_ret_code
- Proxy $\mathcal{A}$ -distance (Fig. 4b of the paper): code/pad_code
- Test-time Inference: code/inference_code
- Plots (Fig. 5, Fig. 6 of the paper): See README.md file
- Trained models (*DomainNet*) / Sample Dataset / Misc.: https://drive.google.com/drive/folders/1Y_8L2tEq6oPIpJqxEsSZNIuQ0ai_qfI8?usp=sharing

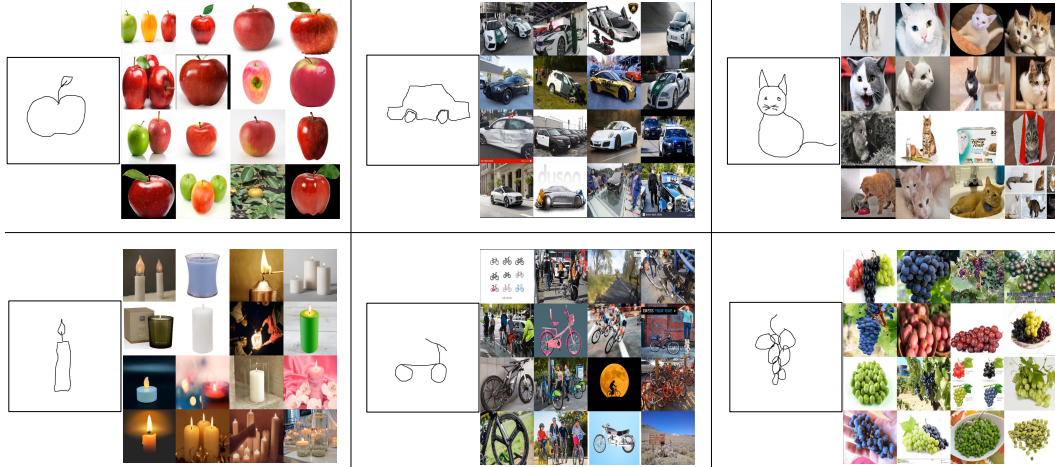


Figure 6: An example showing the top-16 retrievals ($k = 16$ in Line 9 of Algo. 1 above) from the **Rel** domain by manually "doodling" objects (see `image_retrieval_demo.mp4` for the demonstration).

References

- [1] Mingsheng Long, Yue Cao, Jianmin Wang, and Michael Jordan. "Learning Transferable Features with Deep Adaptation Networks". In: *ICML*. 2015.
- [2] Mingsheng Long, Han Zhu, Jianmin Wang, and Michael I Jordan. "Deep transfer learning with joint adaptation networks". In: *ICML*. 2017.
- [3] Mingsheng Long, Han Zhu, Jianmin Wang, and Michael I Jordan. "Unsupervised domain adaptation with residual transfer networks". In: *NeurIPS*. 2016.
- [4] Xingchao Peng, Qinxun Bai, Xide Xia, Zijun Huang, Kate Saenko, and Bo Wang. "Moment matching for multi-source domain adaptation". In: *ICCV*. 2019.
- [5] Kuniaki Saito, Kohei Watanabe, Yoshitaka Ushiku, and Tatsuya Harada. "Maximum classifier discrepancy for unsupervised domain adaptation". In: *CVPR*. 2018.
- [6] Eric Tzeng, Judy Hoffman, Kate Saenko, and Trevor Darrell. "Adversarial discriminative domain adaptation". In: *CVPR*. 2017.
- [7] Ruijia Xu, Ziliang Chen, Wangmeng Zuo, Junjie Yan, and Liang Lin. "Deep cocktail network: Multi-source unsupervised domain adaptation with category shift". In: *CVPR*. 2018.